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Comparison of MRI with EMG to study muscle activity associated with dynamic plantar flexion

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Abstract

This study compared magnetic resonance imaging (MRI) and surface electromyography (EMG) to evaluate the effect of knee angle upon plantar flexion activity in the triceps surae muscles [medial & lateral gastrocnemius (MG, LG) and the soleus (SOL)]. Two weight & height matched groups performed identical protocols, twelve (6M, 6F) in the MRI group, twelve (8M, 4F) in the EMG group. Subjects plantar flexed dynamically for 2 min at 25% of 1-repetition maximum voluntary contraction (1-RM). Exercise was performed with the knee extended (0° flexion), flexed (90°), and partially flexed (45°). In the MRI group spin-echo images were acquired before and immediately following each exercise session. T_2 times, calculated at rest and after exercise by fitting the echoes to a monoexponential decay pattern with a least-squares algorithm, were compared with EMG data. In the EMG group a bipolar electrode was used to collect samples were from the MG, LG, SOL, and anterior tibialis (TA) during exercise at each knee angle, MRI also examined the peroneus (PER). At 0° flexion MRI demonstrated a significant post-exercise T_2 increase in the MG ($p \leq 0.001$), LG ($p \leq 0.001$), and PER ($p \leq 0.01$), with no T_2 change in the SOL or TA. At 90° flexion there was a significant T_2 increase in the SOL ($p \leq 0.001$) with no significant T_2 change in the MG, LG, PER, or TA. At 45° T_2 increased significantly in the SOL ($p \leq 0.001$) and LG ($p \leq 0.05$), but not the MG, PER, or TA. EMG produced similar results with the exception that there was significant activity in the TA during the relaxation cycle of the 90° protocol. We conclude that: 1) Soleus activity is measurable by MRI; and 2) MRI and EMG produce similar results from different physiological sources, and are therefore complementary tools for evaluating muscle activity. © 2003 Elsevier Inc. All rights reserved.

Keywords: MRI; EMG; Motor unit activation; Exercise; Electromyography

1. Introduction

The medial and lateral gastrocnemius (MG & LG) and soleus (SOL) muscles make up a synergistic group known as the triceps surae that actively contributes to human locomotion [1]. Comparisons among the MG, LG, and SOL are interesting because of their common insertion on the posterior surface of the calcaneus via the Achilles tendon [1]. While the triceps surae muscles all plantar flex the foot, they are different anatomically, biochemically, and functionally [2-5]. To maintain a standing posture, the monoarticular SOL is always active countering the effects of gravity [6]. The postural role of the SOL is supported by its mainly Type I fiber composition that is capable of aerobically maintaining prolonged isometric tension [4,6]. The MG and LG have a higher percentage of Type II fibers that can generate the high rates of force development required for more dynamic activities [4,6]. Because the MG and LG originate on the femur, the gastrocnemius can produce more leverage than the SOL [7]. Because they are bi-articular, they are more strongly affected by knee joint angle than the SOL, becoming longer to generate more leverage with the knee extended [7] and shortening as the knee is flexed. Recruitment of the triceps surae is increased as activity

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levels increase from standing to walking to running and jumping [3,6,8].

Historically, electromyography (EMG) has been considered the "gold standard" for studying muscle activity. EMG was recently used to study the functions of the MG and LG, and the SOL muscles during fast running, demonstrating that, when compared to the soleus, EMG activity increased in the gastrocnemius much sooner following ground contact [9]. During the initiation of ground contact the knee shows greater extension than at any other point of the running stride, and it is at full knee extension that the triceps surae are capable of generating the most force [10]. Numerous studies employing EMG have demonstrated selective activity during various tasks involving plantar flexion [2,3,7-9,11,12]. EMG was recently used to demonstrate that muscle activity patterns in the triceps surae are dependent upon knee angle [13]; however, this has not been demonstrated using MRI.

T_2-weighted MR images can be used to measure muscle activity within groups of muscles, which appears as a differential increase in MR properties that is workload dependent [14-17]. These exercise induced signal changes result primarily from increases in the spin-spin (T_2) relaxation time of tissue water [14,16,17]. Therefore, while EMG measures a change in electrical potential occurring along a muscle during muscular contraction, MRI depends on the resulting shift in water brought about by metabolic changes during contraction. MRI has been used to demonstrate activity in the gastrocnemius by plantar flexion [14-16,18-20]; however, activity in the soleus has not been shown with MRI. The advantage of MRI is that it can be used to simultaneously examine large muscle groups that contain many individual muscles. However, in order for the potential benefits of MRI to be realized, its ability to measure changes must be demonstrated in all muscles. The purpose of this study was twofold: 1) to demonstrate a measurable plantar flexion induced signal increase in the soleus using MRI, and 2) to compare the MRI patterns of T_2 increase associated plantar flexion at different knee angles with established [13] EMG activity patterns. We employed MRI and EMG techniques to assess lower leg muscle activity before, during, and after two minutes of repeated plantar flexion's performed at 25% of 1-repetition maximum voluntary contraction (1-RM) and examined the effect of different degrees of knee flexion.

2. Methods

2.1. Subjects

2.1.1. MRI

Twelve healthy subjects (6 male, 6 female) participated in the MRI portion of this study. The MRI subjects were 30 ± 3 years of age, 72 ± 5 kg weight, and 172 ± 3cm in height. Subjects were screened according to the Yale-New Haven Hospital standard clinical criterion for MR imaging. This criterion requires that, when accepted into the study, each MRI subject provided informed written consent in accordance with a protocol approved by the Yale University Human Investigations Committee.

2.1.2. EMG

Twelve healthy subjects (8 male, 4 female) participated in the EMG portion of the study. Subjects were 23 ± 3 years of age, 77 ± 5kg weight, and 173 ± 2cm in height. Each subject provided informed consent in accordance with Human Studies Committee of the University of Massachusetts at Amherst. Circumstances did not permit study of the same group of subjects with both techniques, and this represents a technical constraint of the study. However, subjects in both groups were enrolled under conditions that assumed random selection.

2.2. Exercise protocols

2.2.1. MRI and EMG

Using MRI exercise induced T_2 changes were assessed in the lower leg MG & LG, SOL, anterior tibialis (TA), and peroneus (PER). These muscles were dynamically plantar flexed at a constant and standardized workload of 25% of 1-RM [1-RM = the maximum one-repetition plantar flexion load] for a fixed amount of time (2 min) (60 contractions). Subject 1-RM values were determined at full knee extension (0°) and found to be 290 ± 16 lbs (150–370 lbs) [15,21]. Subjects exercised in a random testing order with the knee flexed at either 90°, 45°, or 0° (0° = full extension) at an angular contraction velocity around the ankle of ~45°/s. Two-minute exercise protocols were performed in the same

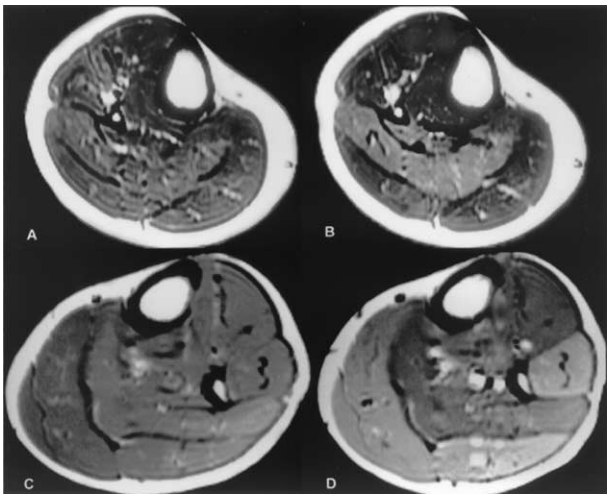

Fig. 1. MRI of the lower leg (mid-calf) obtained before and immediately after exercise. A: Before plantar flexion with the knee flexed at 90°, B: After plantar flexion with the knee flexed at 90°, C: Before plantar flexion with the knee fully extended (0° flexion), D: After plantar flexion with the knee fully extended (0° flexion). At full knee extension the peroneus was recruited to stabilize the ankle.

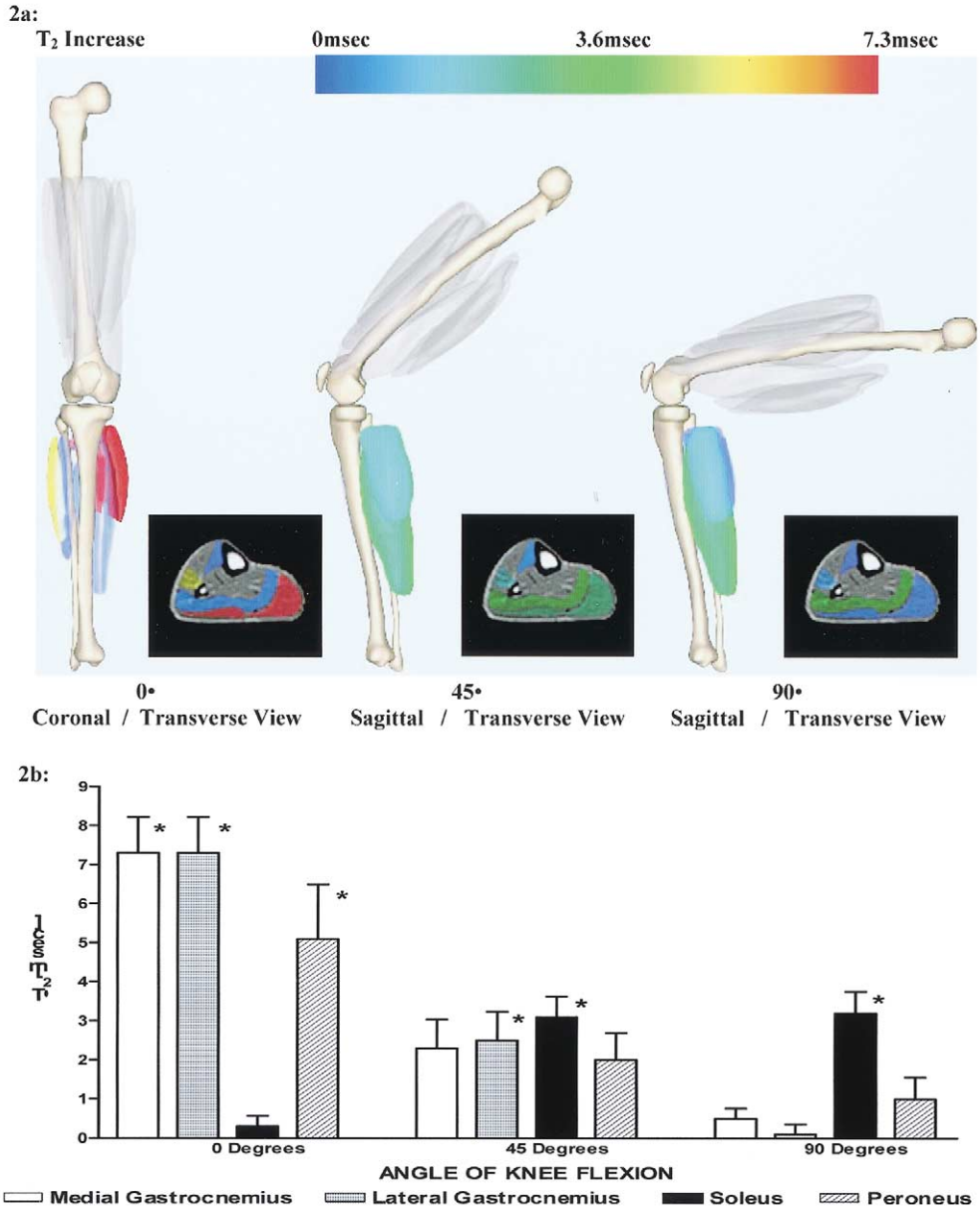

Fig. 2. Plantar flexion induced change in transverse relaxation (T_2) times within the medial gastrocnemius (MG), the lateral gastrocnemius (LG), the soleus (SOL), and the peroneus (PER) at 0°, 45°, and 90° of knee flexion. The tibialis anterior (TA) was not significantly recruited at any of the knee angles tested. 2a. Data are presented as a color image that is a three-dimensional representation of a series of two-dimensional MR images. Colors are related to a scale of T_2 increases that are a direct representation of the degree of metabolic activation of the muscle. MRI typically records exercise-induced T_2 increases as a global change in all MR slices through the muscle; therefore, colors are presented as the same throughout the muscle. Inserts are color overlays onto two-dimensional MR images. 2b. The same data presented as a bar graph. $*p \leq 0.05$ resting muscle versus muscle post-exercise. Values are mean ± SEM.

leg in exercise sessions that were separated by at least one hour. Electrical activity, measured with EMG, ceases immediately at the end of each contraction, while the water movements that are measured with MRI require more time to return to the basal state [16]. During exercise, muscles contracted so as to rotate the ankle through a full range of motion [22] against a constant resistance equal to 25% of 1-RM for the plantar flexion. The average rate of contractions was 30 per minute (~60 total contractions). A single mid-calf MR image (single-slice) was obtained immediately before and after each exercise protocol.

The exercise ergometer was designed to enable dynamic contractions that rotated muscles through a full range of motion against a constant resistance, and has been previously described [15]. Briefly, the ergometer was constructed from non-magnetic materials and mounted to a standard MRI patient table, to permit exercise within the MR imager. To provide constant resistance, a sealed dual-chambered pneumatic cylinder/piston assembly was articulated with a pedal assembly so that the smallest possible angle was subtended during movement through a full range of motion. Air pressure was maintained in a closed-loop between the

pneumatic cylinder chamber and an air supply tank [15]. The system was designed so that when each muscular contraction was released, air refilled the pressurized chamber with no residual resistance from the non-pressurized chamber, actively moving the foot back to a resting position. The pneumatic cylinder/piston assembly was mounted at one end to a rigid Lexan frame and at the other end to a pedal which rotated the foot about the subject's ankle.

In the EMG protocol subjects were seated with their leg placed in the same pneumatic non-magnetic exercise ergometer as used in the MRI protocol. Repetitive plantar flexion contractions (30/min) were performed at a load of 25% of 1-RM. The three conditions (knee angles of 0°, 45°, and 90°) were completed with at least 1hr rest between conditions. The order of administration of the conditions was balanced. The set-up of the exercise ergometer was identical to that used in the MRI protocol, and subjects exercised for 2 min (60 contractions) at an angular contraction velocity of ~45°/s.

2.3. EMG preparation

Pre-amplified high-impedance surface electrodes, 8 mm in diameter (Therapeutics Unlimited) were placed over the belly of the SOL, MG, LG, and TA muscles with a 20 mm center-to-center interelectrode distance. The ground was placed on the lateral aspect of the knee. Signals from these electrodes were bandpass filtered (20Hz-4kHz; −3db), amplified using a differential amplifier, displayed on an oscilloscope, digitally sampled (12-bit resolution, 2000 samples per second), and stored for later analysis. The same electrodes remained in place for all three conditions (0°,45°,90°). A preliminary cross-correlation analysis revealed negligible crosstalk among these EMG channels.

2.4. EMG analysis

The EMG signals were acquired continuously throughout the two-minute exercise protocol. The exercise ergometer (pneumatic device) and exercise performance parameters (angular contraction velocity = 45°/sec) were identical to those of the MRI protocol. During contractions the root mean square of the raw EMG signal (rmsEMG) was used to quantify the level of electrical activity for each muscle. Variations in electrode area, quantity and conductance of the gel, surface impedance, daily changes in amplifier characteristics, and temperature (23–25) can influence EMG signal. Given the repeated measures design employed within a single test day and the balanced treatment assignment, these modifying factors were considered to be constant within each subject's testing session.

2.5. MR imaging

According to the designed exercise protocol, T_2 times were compared before and after each of the three exercise sessions at 0°, at 45°, and at 90°. This data collection protocol is different from EMG data, which was collected as electrical activity during muscular contraction, and not before and after exercise. Single slice MR imaging was performed on a 1.5-Tesla GE Signa system (General Electric, Milwaukee, WI). Subjects were positioned supine within the magnet with an extremity coil positioned midcalf and with their foot positioned in the pedal assembly of the exercise apparatus [15,16]. Transverse mid-calf images (10 mm slice thickness; FOV = 20 cm; 128 × 256 matrix; NEX = 1) were obtained using a multiple spin-echo sequence (TR = 1000 msec; TE = 30,60,90,120 msec; total scan time = 2min 20sec). T_2 values before and after exercise were calculated in multiple regions-of-interest within the MG, LG, SOL, TA, and PER muscles. Each ROI was selected so that visible blood vessels and fat were avoided. Based upon multiple ROI's, the T_2 change was found to be uniform within each muscle tested. T_2 values were calculated by fitting four data points (4 echoes) to a mono-exponential decay using a least squares algorithm.

2.6. MR image preparation and presentation

To generate images that would allow T_2 data to be presented as a three-dimensional color map of the lower leg, two-dimensional MR images were processed using a digital segmentation and reconstruction process. The intention of this type of presentation was to allow readers to get a more global sense of the changes in muscle activity patterns that result from altering the knee angle. Selected muscles and bones in transverse serial MRI slices taken from a representative test subject were digitally traced and shaded in cross-sections using Photoshop 5.0 (Adobe Systems, San Jose, CA). The resulting segmented sections were stacked and reconstructed into polygonal models via a marching cubes algorithm, using Synu graphical software (National Center for Microscopy and Imaging Research at San Diego). The reconstructed muscles and bones were imported into the digital modeling and animation software 3D Studio Max (Kinetix, Milpitas, CA), in which muscles were assigned false colors according to their ΔT_2 values. A linear color scale of indigo through red (hues of 0°–170° in 3D Studio Max HSB color space) was chosen to represent the measured range of ΔT_2 values of 0 (indigo) to 7.3 (red) at knee flexion angles of 0°, 45°, and 90°. Any negative ΔT_2 values were not significantly different from zero and were taken to represent no significant change in T_2. They were therefore assigned a value of zero. The reconstructed bones were assigned a pivot point at the position of the knee joint and the entire reconstructed leg flexed from its initial rest angle of 0° (straight), to simulate 45° and 90° bends for the purpose of illustration (Fig. 2a). Two-dimensional MR images are presented as inserts in Fig. 2a. These were overlaid with the same color scale as the three-dimensional reconstructions.

Table 1
Mean T_2 ± SE before and after 2 min plantar flexion exercise at 0°, 45°, and 90° of knee flexion

KNEE-ANGLE	STATUS	MG T_2 [msec]	LG T_2 [msec]	SOL T_2 [msec]	PER T_2 [msec]	TA $_2$ [msec]
90°	REST	29.4 ± 0.4	29.5 ± 0.4	31.1 ± 0.4	29.7 ± 0.5	29.2 ± 0.8
	EXE.	29.7 ± 0.5	29.6 ± 0.5	34.3 ± 0.5**	30.7 ± 0.4	27.9 ± 0.5
45°	REST	30.0 ± 0.6	30.0 ± 0.6	31.1 ± 0.5	30.54 ± 0.8	28.0 ± 0.4
	EXE.	32.3 ± 1.0	32.5 ± 1.0†	34.2 ± 0.8*	32.4 ± 1.0	28.6 ± 0.8
0°	REST	30.5 ± 0.7	30.9 ± 0.8	31.1 ± 0.7	30.9 ± 0.9	29.9 ± 1.8
	EXE.	37.8 ± 1.1**	38.2 ± 1.1**	31.4 ± 0.9	36.0 ± 1.7†	29.4 ± 1.0

Comparisons are made within each muscle, between resting and post-exercise T_2 times: † $p \leq 0.05$, * $p \leq 0.01$, ** $p \leq 0.001$. MG = medial gastrocnemius, LG = lateral gastrocnemius, SOL = soleus, PER = peroneus, TA = tibialis anterior.

2.7. MRI and EMG statistical analyses

EMG amplitudes were computed for the duration of each of the 50 repetitions at each knee angle, averaged, and used for statistical analyses. Repeated measures analysis of variance (ANOVA) was used to detect significant differences among different muscles at each knee angle tested and in the same muscle at different knee angles. When significant angle × muscle differences were detected, individual ANOVA was performed to determine individual differences, followed by a Tukey's Studentized Range Test to ascertain the source of the difference. All EMG statistical analyses were performed using the PC-SAS statistical analysis package. Values of T_2 in exercised muscles (MG, LG, SOL, PER, & AT) at rest and at 1.16 min following exercise were compared in a before and after fashion using ANOVA followed by a post-hoc comparison of means (Bonferroni).

3. Results

3.1. MRI

Fig. 1a–d presents MR images that were acquired before and immediately after (2 min) plantar flexion at 90° and 0°. Following exercise, the muscles that were recruited appear hyper-intense on these T_2-weighted images (1b & 1d). In Table 1, mean T_2 times are given at rest (before exercise) and immediately after exercise (within 2 min), while Fig. 2a and 2b shows calculated increases in T_2 times at the three different knee angles. Following exercise at 0° T_2 times in the MG and LG increased significantly ($p \leq 0.001$,MG; $p \leq 0.001$, LG) relative to baseline T_2 times, indicating that these muscles were recruited and performed work during plantar flexion. There was no increase in the SOL. As there was no dorsi-flexion component to the exercise protocol, T_2 times in the TA did not increase. MRI was able to measure T_2 times in the peroneus (PER) observing a significant increase ($p \leq 0.02$), and suggesting that at 0° this muscle was recruited to stabilize the ankle during plantar flexion (Table 1, Fig. 1d and Fig. 2). At 45° T_2 times in the LG and SOL increased ($p \leq 0.05$, LG; $p \leq 0.01$,SOL) following exercise, but T_2 times in the MG were not significantly increased. However, while significance was not established in the MG, the muscle did appear to be at least marginally activated (Table 1, Fig. 2b). There were no significant increases in the TA or the PER at 45°; although, as in the MG, there did appear to be marginal activity of the PER. At 90°, only T_2 times in the SOL were significantly increased ($p \leq 0.001$) following exercise (Table 1, Fig. 2b).

3.2. EMG

Mean amplitudes for electrical activity of the medial gastrocnemius (MG), lateral gastrocnemius (LG), soleus (SOL, and tibialis anterior (TA) are pictured in Fig. 3a-3d. A repeated measures analysis of variance, incorporating the three different knee angles (0°, 45°, & 90°) as a repeated measure, was conducted within each muscle. Fig. 3c illustrates that there were no statistically significant differences in EMG amplitude among the three conditions for the SOL muscle ($F = 0.07$, $p < 0.9293$). However, the three different knee angles produced significant differences in rmsEMG amplitudes for the LG ($F = 8.1$, $p < 0.0001$), MG ($F = 14.3$, $p < 0.0001$), and TA ($F = 14.3$, $p < 0.0001$) among the three conditions (Fig. 3a,b, & d). For the LG and MG, Tukey's range test revealed that EMG amplitudes for the 45° and 90° conditions were similar but significantly different from the 0° knee joint angle ($p < 0.04$). For the TA, the EMG amplitudes were similar at the 0° and 45° angles, but significantly different at the 90° joint angle ($p < 0.004$). Fig. 3a & 3b illustrates that considerably greater EMG activity existed when the knee was straight (0°) than when it was bent. The antagonist TA activity was small at 0° and 45°, but increased considerably at the 90° joint angle (Fig. 3d).

4. Discussion

In this study, MRI was used to assess muscle activity in the lower leg following plantar flexion exercise performed

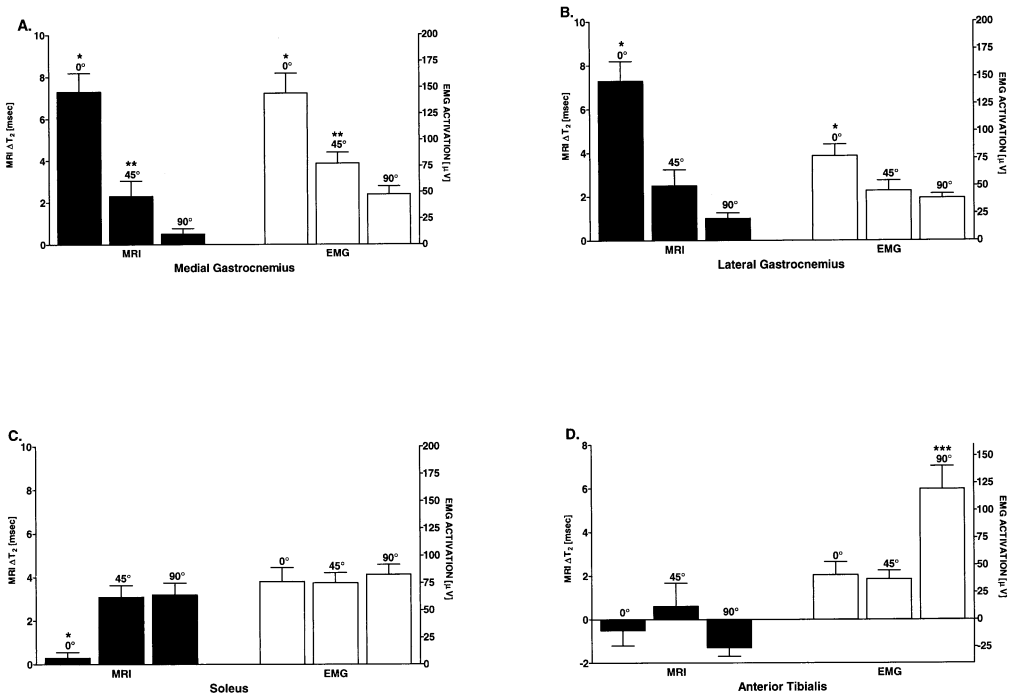

Fig. 3. Plantar flexion induced MRI T_2 changes [msec] and EMG activation [μV] of the MG, the LG, the SOL, and the TA at 0°, 45°, and 90° of knee flexion. 3a. MG T_2 changes MRI and EMG activation are greater at 0° than at 45° and 90° (*$p < 0.006$). MRI and EMG are greater at 45° than at 90° (**$p < 0.04$). 3b. LG T_2 changes MRI and EMG activation are greater at 0° than at 45° and 90° (*$p < 0.04$). 3c. SOL T_2 changes MRI is less at 0° than at 45° and 90° (*$p < 0.0003$) but not significantly different for EMG. 3d. TA EMG activation was greater at 90° than at 0° and 45° (***$p < 0.004$) but not significantly different for MRI. Negative values for TA MRI were not significantly different from zero. All values (3a-d) are mean ± SEM.

with 0°, 45°, and 90° of knee flexion. MRI data were compared with surface EMG data of the same muscle during the same exercise. MRI and EMG scores were compared on the basis of their ability to detect qualitatively similar patterns of muscle activity. EMG and MRI both demonstrated progressively less activity of the MG and LG as the leg was flexed; however, there was less agreement between MRI and EMG in the analysis of SOL activity. In the EMG study there was considerable activity of the SOL at all knee angles, while MRI did not detect activity in the SOL when the knee was fully extended. It is likely that this discrepancy results from the different physiological phenomena that are detected by EMG and MRI. The MRI analysis unexpectedly detected activity in the PER, a muscle not targeted for study using EMG. In this study the PER is believed to have stabilized the ankle; which, because the ankle was not supported by the exercise apparatus, required increasing stabilization as the knee angle moved from 90° to 0° of flexion. Activity of stabilizing or postural muscles during voluntary contraction has been well-documented, and such muscles can begin firing even prior to the prime movers involved in the task [26]. In the TA, an antagonist of the triceps surae, the T_2 did not increase and the EMG activity increased only during the return phase of the exercise protocol while the knee was flexed at 90° (Fig. 4). This observed EMG activity at the 90° knee angle may have resulted from the design of the exercise ergometer, which allowed the ankle to move through the primary range-of-motion of dorsi-flexion in the 90° knee orientation initiating a co-contraction of the TA with the SOL. This would have resulted in a "braking-action" during the relaxation phase of each contraction/relaxation cycle [27] that would have manifest as EMG activity in the TA. No activity of the TA was detected with MRI at any time during this study, suggesting that EMG activity in the TA at 90° was not necessarily associated with the muscle performing work.

The mechanisms by which physiological events initiate a change on MR images is complex, and at present are not fully understood. It is thought that exercise-induced T_2 increases are driven primarily by water shifts resulting from transient changes in tissue osmolality and perfusion [28,29] consequent to metabolic activity, though perfusion is not necessary for exercise-induced T_2 increases [18,30]. This suggests that accumulation of tissue osmolytes, particularly intracellular lactate, may be an important factor effecting T_2 increases [31-33]. These early studies also reported that when blood perfusion was returned to muscle tissue following exercise, there was an additional increase in T_2, suggesting that osmolality and perfusion may be additive [18,30]. It is likely that exercise induced decreases in intracellular pH also plays a role in the T_2 of muscle [31].

The T_2 changes following exercise, which scale to the intensity of the work performed [14,15], are therefore sensitive to the metabolic activity required to generate force. EMG measures the wave of depolarization that initiates muscle contraction, and is sensitive to the recruitment of

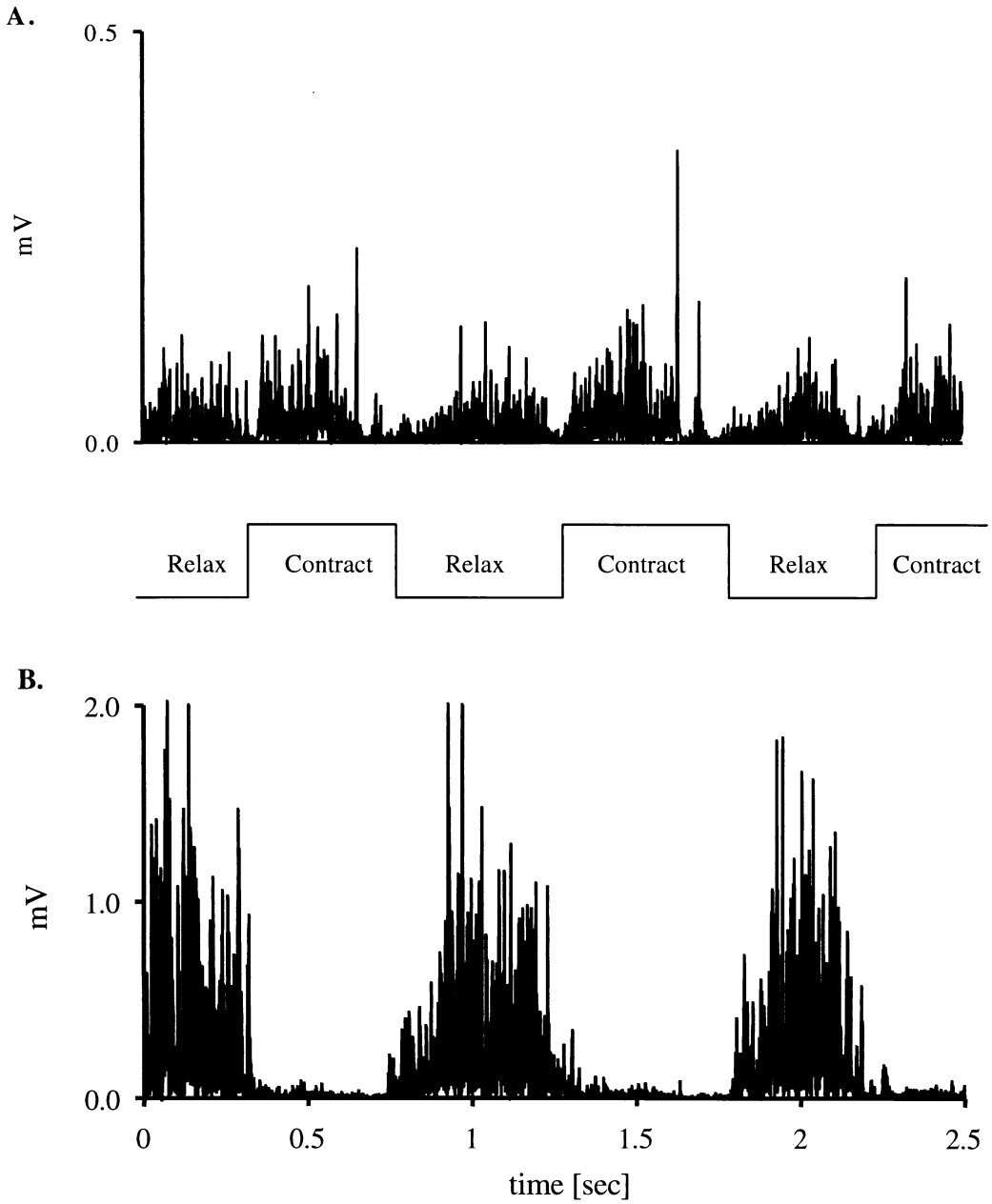

Fig. 4. Two and a half seconds of myoelectric activity at 90° of knee flexion from: (A) the soleus and (B) the tibialis anterior of an individual subject. Unlike the other knee orientations (0° and 45°, data not shown), at 90° the behavior of these two muscles was different. While there were soleal plantar-flexor bursts at 0.5 sec, 1.5 sec, and 2.5 sec following the initiation of the contraction, there was very little tibialis anterior activity. During the relaxation phase (at 0 sec, 1 sec, and 2 sec), tibialis anterior was active and co-contracting with the soleus.

motor units and to the rate and patterns with which they fire. It therefore reflects the strategy employed by the nervous system to generate force or to execute a movement. The absence of T_2 changes in the presence of EMG activity suggests that although the muscle was recruited by the nervous system, it did not perform enough metabolic work to undergo a T_2 change. The relevant cases in the present study occurred at full extension, where there was EMG activity but no T_2 change in the SOL, at 90° of flexion, where EMG activity but no T_2 change was present in the TA, and at 90° where there was EMG activity in the LG and MG but no T_2 change. In the former case, this may have occurred because the contribution of the LG and MG to force production dominated that of the SOL. In the case where LG and MG activity appeared in the absence of T_2 change, this may have occurred because the bi-articular gastrocnemius was moved passively to a position on the length-tension curve at which there was little opportunity for cross-bridge cycling [5,34,35]. As a result, there would be less metabolic activity creating a reduced demand for

ATP. The overlap of thin and thick filaments in the monoarticular SOL, conversely, is not affected by knee flexion, and so at this knee angle it must dominate the force production of plantar flexion.

By detecting selective activity of the muscles of the lower leg during plantar flexion at different knee orientations, this study demonstrates the efficacy of MRI for simultaneously detecting activity in groups of muscles. This is the first report of the use of MRI to detect activity in the soleus during plantar flexion, and it is an essential step in the development of MRI as a tool to study the muscle activity patterns that result from complex movements. In order to be able to evaluate unknown muscle activity patterns the need exists to be able to think of different skeletal muscles as being basically similar in their MRI behavior. This study, by using MRI to examine a known movement (plantar flexion) that produces a known muscle activity pattern [2,3,11,16,17,36,37], demonstrates that MRI is a viable tool for muscle activity studies.

The current MRI and EMG data demonstrate that the relative contributions of the triceps surae muscles during plantar flexion are a shared phenomenon. The muscles of the triceps surae are utilized during plantar flexion according to knee angle and the corresponding changes in muscle length. The current EMG results are in agreement with those reported by Tomaki et al. [13] comparing the MG, LG, and SOL at knee angles of 0°, 30°, and 60° at speeds of 6°/sec, 30°/sec, and 60°/sec, and relative loads of 5% and 10% of 1-RM. That study reported significantly higher peak integrated EMG (IEMG) values in the SOL at 60° than at 30° and 0°. The opposite pattern was reported for the MG, with IEMG values being significantly higher at 0°. The study concluded that during plantar flexion the effect of knee angle upon muscle activity in the triceps surae was the dominant variable. Preliminary work in our laboratories has examined EMG patterns at 382°/sec observing, in agreement with Tomaki et al. [13], that the effect of knee angle was dominant over the contraction velocity (unpublished observation). However, the potential effects of other factors upon lower leg muscle recruitment during plantar flexion must not be ignored. Other studies have reported differences in muscle recruitment patterns due to fiber type content and force of contraction [36,38]. In this study the results of EMG analyses of the MG, LG, and SOL muscles during plantar flexion reveal notable differences in muscle utilization patterns due to knee angle. These patterns can also be explained by noting the monoarticular structure of the SOL and the biarticular structures of the MG and LG. Since the MG and LG cross both the knee and ankle joints, their length and relative tension during plantar flexion would be affected by changes in the knee angle. Because the SOL has its origin on the soleal line of the tibia and posterior head and upper shaft of the fibula, changes in knee angle would be expected to have little effect on its function, except as it relates to its functional interaction with the MG and LG.

In summary, this study demonstrates that EMG and MRI are both capable of detecting patterns of muscle activity among the leg muscles during plantar flexion exercise. However, the results of the two techniques were not in complete agreement. This is because EMG is sensitive to the electrical activity of a muscle, while changes in T_2 are brought by metabolic activity within the muscle, pointing to potential advantages and disadvantages of each technique. Surface EMG, while rapidly responding to electrical changes in the underlying muscle fibers, can only detect the activity in those surface muscles for which it has been specifically targeted. Inasmuch as the surface EMG signal is biased by the activity of more superficially oriented muscle fibers [39], MRI changes, which respond more slowly to changes in muscle activity, are able to detect the metabolic activity of deeper muscle fibers and possibly other muscles. The two techniques may therefore be viewed as complementary, rather than competitive. The plantar flexion activity patterns detected by EMG agree with those reported in other studies [2,3,11,16,17,36,37]. The relative agreement between MRI and EMG observed in this study suggests that MRI is a valid tool for studying muscle activity. From these data, we conclude that muscle activity during plantar flexion is a shared phenomenon, with the degree of contribution of the plantar-flexor muscles being influenced by the degree of flexion of the knee joint. We further conclude that this phenomenon is measurable by both EMG and MRI [23-25].

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